Department of Clinical and Molecular Medicine

Examination paper for
MOL3100 - Introduction to Molecular Medicine with project
MOL3000 - Introduction to Molecular Medicine

Examination date: 08.12.22

Examination time (from-to): 15.00-19.00

Permitted examination support material: D (No printed or hand-written support material is allowed. A specific basic calculator is allowed.)

Academic contact during examination: Bodil Kavli, Phone: 97661442

OTHER INFORMATION
This exam has two parts. Part 1 essay questions (Q 1-7) where you write your answer in the textbox and Part 2, multiple choice questions (MCQ 1-25). Each part counts 50% of the exam.

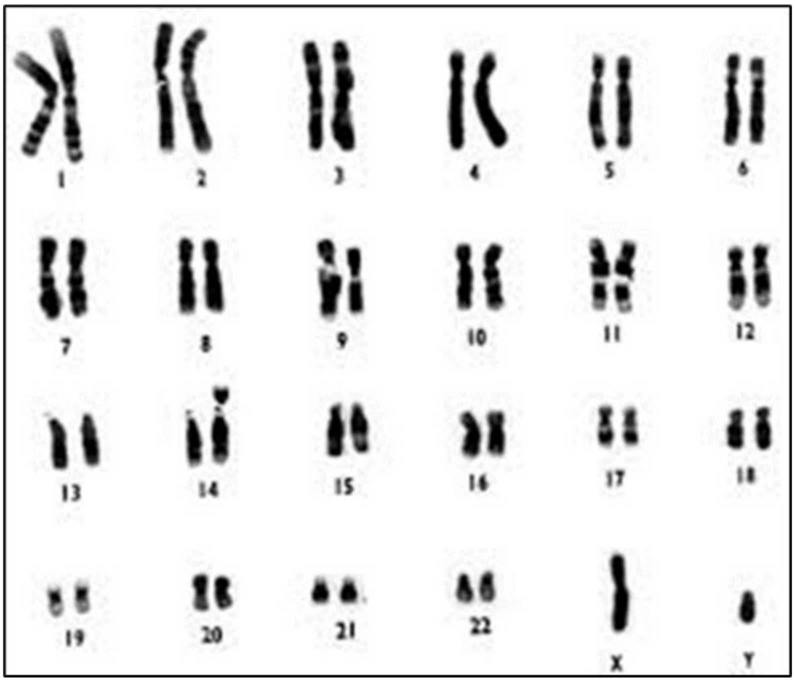
Make your own assumptions: If a question is unclear/vague, make your own assumptions and specify them in your answer. Only contact academic contact in case of errors or insufficiencies in the question set. Address an invigilator if you wish to contact the academic contact. Write down the question in advance.

Notifications: If there is a need to send a message to the candidates during the exam (e.g., if there is an error in the question set), this will be done by sending a notification in Inspera. A dialogue box will appear. You can re-read the notification by clicking the bell icon in the top right-hand corner of the screen.

Withdrawing from the exam: If you become ill or wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Essay Q1 (2P)

A newborn boy was diagnosed with Down syndrome. His karyotype is shown below.



Explain why the karyotype is indicative of Down syndrome and what has most likely happened.

Answer:
Down syndrome is caused by trisomy 21. Apparently the boy has only two copies of intact chromosome 21, so the third copy has to be attached to another chromosome. It can be seen from the karyotype that one of the chromosome 14 copies is somewhat larger than the other copy, suggesting that the third copy of chromosome 21 is attached to chromosome 14. Chromosome 21 and 14 are both acrocentric chromosomes and can be subject to Robertsonian translocations. In Robertsonian translocations between acrocentric chromosomes the chromosomes break at (or near) the centromeres and the long arms fuse to form a single, large chromosome. In this case most of chromosome 21 is attached to chromosome 14, causing trisomy 21 and Down syndrome.
The child might have inherited the fusion chromosome in two ways. 1. One of the parents carry this chromosome in all cells (including germ cells) but is phenotypically normal since there is no loss of critical genetic material (the short arms contain mostly rRNA genes which are also found on other acrocentric chromosomes). 2. The translocation has occurred in the germ cells of one of the parents.

Essay Q2: (2p)
Chromosomal translocation is a term used to describe the interchange of parts between two non-homologous chromosomes. Sometimes chromosomal translocations are considered the primary cause for cancer, including lymphoma, leukemia, and some solid tumors.

Explain how a translocation event may lead to cancer and give a relevant example.

A relevant example (and presented to the students in the lecture about mutations) of such an event is the t(9;22) translocation (Philadelphia chromosome), which results in the regulation of ABL gene on chromosome 9 by the BCR gene promoter on chromosome 22 due to the formation of a unique in-frame fusion mRNA and protein. The protein coded for by the abnormal BCR-ABL1 fusion gene is a tyrosine kinase with enhanced activity, which is responsible for the uncontrolled growth of leukemic cells in patients with chronic myelogenous leukemia (CML), and in some patients with acute lymphoblastic leukemia (ALL) or acute myelogenous leukemia (AML). (Other examples are acceptable if relevant. E.g. translocation placing the MYC proto-oncogene from chromosome 8 under the control of the powerful immunoglobin heavy chain gene (IGH) promoter on chromosome 14. The MYC protein normally signals for cell proliferation, and the translocation causes high levels of MYC overexpression in lymphoid cells, where the IGH promoter is normally active. This example has not been lectured)

Essay Q3: (2p)
Refer to the following table, which shows the sequence of amino acids (asp, gln, gly, pro, and tyr) in the corresponding section (amino acid position 291-300) of a given protein (X) in two individuals (A and B).

Individual	Amino acid position, protein X									
	291	292	293	294	295	296	297	298	299	300
A	pro	gln	gly	asp	tyr	gly	gln	gln	gly	tyr
B	gly	pro	gln	gly	asp	tyr	gly	gln	gln	gly

Explain two possible mutational events that can be responsible for the differences in the amino acid at the given positions in protein X in individuals A and B

Answer: Either an <u>in-frame deletion (one codon three bases)</u> upstream of codon 291 in individual A, or an <u>in-frame insertion (one codon)</u> at or upstream of codon 291 in individual B

Essay Q4: (6p)

Apoptosis (programmed cell death) is a normal physiological mechanism in multicellular organisms important in development and cell homeostasis.

A. List at least five characteristics of apoptotic cells. (1p)
B. Three pathways of apoptosis were mentioned in the lecture. Briefly describe these pathways (main features). (3p)
C. In what way can insufficient and excessive apoptosis be involved in disease, mention a few examples. (2)

Answer

a)

- the cell shrinks
- the cell develops bubble-like blebs on their surface (blebbing)
- the chromatin (DNA and protein) condensate
- the chromatin in nucleus degrades (DNA fragmentation)
- the mitochondria break down with (usually) the release of cytochrome c
- the cell break into small, membrane-wrapped, fragments
- the phospholipid phosphatidylserine, which is normally hidden within the plasma membrane, is exposed on the surface (Annexin positive= apoptotic cells)

b)

1. Extrinsic pathway: killing after external signalling. Direct cleavage of caspase 8 or 10 by ligand binding to (fas/fast ligand)/activation of receptors. Caspase 8/10 next cleaves procaspase 3/7

2. Intrinsic pathway: killing after internal stress. DNA damage or absence of growth factors alters the balance between apoptotic and anti-apoptotic proteins leading to release of cytochrome C from mitochondria (the mitochondrial pathway). This activates the Apoptosome and cleave pro-caspase 9, which then cleaves procaspase 3/7

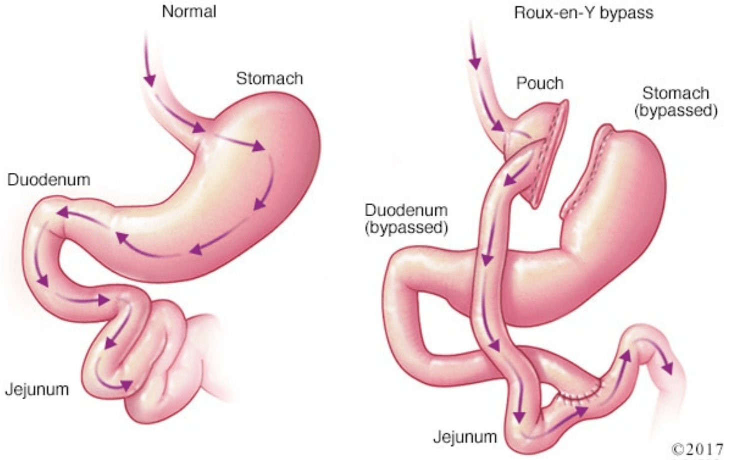
3. Pro-caspase binding to PCNA, inhibition of this binding via alterations of PCNA (during neutrophil maturations) or targeting of PCNA induce caspase 8, 9, 3 and 7 cleavages.

c)

- Insufficient apoptosis (cancer cells, infected cells)
- Excessive apoptosis (neurodegenerative diseases, tissue damage after stroke)

Essay Q5: (4p)

The prevalence of type 2 diabetes has increased markedly in parallel with obesity. Bariatric surgery (Roux-en-Y bypass) is known to induce not only bodyweight reduction but also type 2 diabetes (T2D) remission.



A. What is the current understanding of the mechanism behind this surgery-induced T2D remission? (2p)

B. Describe how we can use this knowledge in development of molecular medicine for T2D? (2p)

A. Answer: Due to changes in anatomy by the surgery, ingested food reaches rapidly the small intestine where L cells are located, leading to an exaggerated release of GLP-1 and PYY, and then GLP-1 and PYY stimulate the beta-cells in pancreas, leading to insulin release.

B. Answer: Potential medicines for T2D may include molecules that mimic GLP-1 and PPY. Since GLP-1 has a short half-life (1.5 – 5 min) due to dipeptidyl peptidase-4 (DPP4), DPP4 inhibitors can be considered for treatment of T2D (In addition, new research indicated that "inactive GLP-1 (9-36)" might have the incretin effect too).

Essay Q6: (6p)

Cancer is a diverse group of diseases that can be defined by a common set of hallmarks. Many different therapeutic strategies and drugs are available to treat various forms of cancer.

A. Describe the essential differences between chemotherapy, targeted therapy, and immune therapy. (3 p)

B. List the main limitations of each treatment strategy (3 p)

Answer:

A) Difference

- Chemotherapy targets all cells, rapid proliferating cells more sensitive
- Targeted therapies aim to target specifically pathways /proteins mutated/altered in cancer cells
- Immunotherapies aim to help the immune system to recognize /kill cancer cells

B) limitations

- Chemo: Side effects due to strong effect on healthy proliferating tissues, epithelial cells, hematopoietic cells etc. Induce a lot of mutations, are strongly mutagenic,- thus, increase chance of secondary cancer.
- Targeted: rapid development of resistance, also side effects on healthy tissue. Reprogramming of signaling pathways both via mutations and epigenetic changes. Signaling network rather than pathways.
- Immunotherapy: clinical benefit only for a fraction of patients

Check point: acquired resistance because of immune evasion (i.e., expression of check point receptors on both immune cells, other cells in TME, and tumor cells), low neoantigen levels in cancer cells, immune related toxicity due to autoimmunity/inflammation, cytokine storm, Check point /CAR-T: price, not for all parts of the world

Essay Q7: (3p)

There are two types of brain strokes with different etiologies and treatment options.

A. Name and describe the major differences between the two types of brain stroke and highlight possible treatment strategies for each type. (2p)

B. The circle of Willis is a junction of arteries at the base of the brain. Describe why this anatomical structure is important in protecting the brain. (1p)

A. Answer:

1. ischemic stroke will happen when there is something (usually blood clot or other type of blockage) clogging a blood vessel

2. hemorrhagic stroke will happen when there is a bleeding in the brain (due to eg a rupture of an aneurysm)

treatment

1. ischemic stroke: thrombolysis using clot dissolving medicine, or endovascular treatment/thrombectomy which make use of a catheter to remove the blood clot

2. hemorrhagic stroke: stop the bleeding, which may involve surgery repair of aneurysm

B: Answer:
The circle of Willis provides collateral blood flow between the anterior and posterior circulations of the brain, protecting against ischemia in the event of vessel disease or damage in one or more areas.

MCQ (1p each)
Only one letter (A, B, C or D) for each question (the best alternative).

1.

The human haploid genome contains ~3 x 10^9 base pairs.	
How many base pairs in total represent genes and unique non-repetitive DNA in a single somatic cell?	
A	~3 x 10^9
B	~6 x 10^8
C	~3 x 10^8
D	~1.5 x 10^9

2.

Using labeling techniques and flow cytometry, the cell cycle phase of each cell in a cell culture can be determined.	
Which biological molecule is usually labeled and quantified per cell to study the distribution of cells in the cell cycle?	
A	RNA
B	Protein
C	DNA
D	Glucose

3.

DNA is an unstable molecule that must be maintained by several DNA repair mechanisms to preserve genome stability.	
Most DNA lesions are caused by	
A	UV and ionizing radiation
B	Endogenous sources
C	Genotoxic agents in the environment and food
D	Chemotherapy

4.

The human diploid genome contains ~6 x 10^9 base pairs.	
How many base pairs represent coding exon sequences in a single germ cell?	
A	~1 x 10^9
B	~3 x 10^8
C	~1 x 10^8
D	~3 x 10^7

5.

The cell cycle is divided in phases that reflects the major cellular processes that occur at each phase.	
What is the order of the cell cycle phases?	
A	G1, S, G2, M
B	G1, G2, M, S
C	S, G1, M, G2
D	S, G1, G2, M

6.

Using labeling techniques and flow cytometry, the cell cycle phase of each cell in a cell culture can be determined.	
Which phase of the cell cycle will represent the highest number of cells after analyzing freely cycling cells?	
A	G1
B	G2
C	S
D	M

7.

DNA methylation is the best studied epigenetic mark and occurs on cytosine at CpG sites.	
In which DNA elements are the CpG sites usually unmethylated	
A	CpG islands
B	Transposable elements
C	Gene bodies (exons and introns)
D	Repetitive elements

8.

Apoptosis is programmed cell death driven by caspases	
Inactive procaspases are activated by	
A	Dimerization and auto-phosphorylation
B	Dimerization
C	Dimerization and auto-proteolytic cleavage
D	The proteasome

9.

PCNA (proliferating cell nuclear antigen) is highly expressed in proliferating cells	
What is the role of PCNA in DNA replication?	
A	PCNA is a helicase
B	PCNA protects single stranded DNA
C	PCNA removes the RNA primer
D	PCNA coordinates the replication machinery

10

Irinotecan, Etoposide and Doxorubicin are widely used in cancer treatment	
What is the target molecule for this class of inhibitors?	
A	DNA polymerase alpha complex
B	Topoisomerase
C	PCNA
D	CMG helicase complex

11

Non-coding RNAs (ncRNAs) can have different roles in disease.	
Which of the following statements is the most accurate in describing ncRNAs in disease?	
A	Most miRNAs are oncogenes
B	LncRNAs have little diagnostic potential
C	ncRNAs are mostly epigenetic markers of environmental exposure
D	Many disease-associated genetic variants are found in non-coding elements

12

Long non-coding RNAs' (lncRNAs) possible molecular functions depend on their cellular localization.	
Which of the following functions is most likely for a nuclear lncRNA?	
A	Codes for a small peptide
B	Acts as a guide RNA for protein
C	Involved in signaling
D	Acts as a microRNA sponge

13

Single nucleotide polymorphisms can affect the function of non-coding RNAs.	
Which of the following SNPs is likely to have the strongest effect on a microRNA?	
A	An SNP in the microRNA promoter
B	An SNP in the microRNA target site
C	An SNP in the hairpin-loop of the microRNA precursor
D	An SNP in the seed region of the mature microRNA

14

In cancer cells the strict control of the cell cycle is usually disturbed due to dysregulation of critical factors.	
Which statement below is most true?	
A	Cyclins are often downregulated in cancer
B	Cyclin-dependent kinases (CDKs) are often downregulated in cancer
C	CDK inhibitors are often upregulated in cancer
D	CDK inhibitors are often downregulated in cancer

15

DNA methylation status in human cells varies during embryonic development.	
What is on average the genomic methylation grade (in % of CpG sites in the genome) in differentiated somatic cell?	
A	<10%
B	~20-40%
C	~60-80%
D	>90%

16

Active DNA demethylation is dependent on TET (Ten-Eleven Translocase) enzymes	
Which chemical reaction is catalyzed by these enzymes?	
A	Excision of the 5-methyl group on 5-methylcytosine
B	Excision of the methylated cytosine base
C	Oxidation of cytosine in position 5
D	Oxidation of the methyl group on 5-methylcytosine

17

BLAST is a popular tool to perform sequence alignments.	
What is a common motivation to perform protein sequence alignment?	
A	Identify Regulatory Elements
B	Discover Protein-Protein Interactions
C	Create 3D Phylogenetic footprints
D	3D Homology modelling

18

Sequence alignments are often calculated using a scoring matrix.	
Which of the following statements correctly describes the elements in a scoring matrix?	
A	The elements are either 1 or 0, depending on whether amino acids are identical or not
B	Elements along the diagonal always has a constant high score, since these positions represent identical amino acids
C	Elements in scoring matrixes can only be created for proteins
D	The elements are scores, representing similarities of amino acids

19

Different stem cell types are defined by their potential to differentiate.	
Totipotent cells	
A	can differentiate into all cell types from three germ layers
B	can differentiate only into components required for development
C	have differentiation potential lower than pluripotent stem cells
D	can differentiate only into components of the same germ layer

20

Type 1 diabetes (T1D) starts with genetic predisposition in combination with external triggers to induce auto-immune reaction and progresses with loss of beta-cell mass.	
How do we best estimate the beta-cell mass in patients?	
A	Plasma glucose level
B	Plasma HbA1c level
C	Plasma C-peptide level
D	Plasma insulin level

21

Robertsonian translocations generally involve the five acrocentric chromosome pairs 13, 14, 15, 21 and 22.	
What characterizes an unbalanced Robertsonian translocation?	
A	Both chromosomes involved contribute with unbalanced amounts of genetic material
B	There is no net loss of genetic material – but phenotypically affected
C	There is net loss of genetic material – but in general phenotypically normal
D	The translocation is unbalanced around the centromeres of both chromosomes involved

22

The N-terminal tails of Histones are usually covalently modified.	
Which amino acid is the substrate for histone acetyltransferases?	
A	Serine
B	Tyrosine
C	Lysine
D	Arginine

23

Monogenic dominant diseases can follow an autosomal or X-linked inheritance pattern. In this task, the disease has full penetrance.	
Which of the following observations points most strongly towards autosomal dominant inheritance?	
A	A sick mother has a healthy daughter and a sick son
B	A sick father has a healthy daughter and a healthy son
C	A sick father has a healthy son and a sick daughter
D	A sick mother has a healthy son and a sick daughter

24

A person diagnosed with hypercholesterolemia was found to harbor an inactivating nonsense mutation in the LDL-receptor gene on one chromosome, but no mutation in this gene on the second chromosome copy.	
What is the most likely explanation for this finding?	
A	The person most likely harbors a second mutation in a different gene
B	The diagnosis is most likely incorrect
C	Mutation in one gene copy is sufficient
D	The person is most likely homozygous for the mutation

25

Aptamers are useful in therapeutics and diagnostics.	
How do aptamers work?	
A	By downregulating protein expression at a post-transcriptional level
B	By silencing transposons
C	By binding specific ligands with high affinity through steric interactions
D	By cleaving transcripts